Run 3 $H + c\ (H \to WW \to 2\ell 2\nu)$

HWW Meeting

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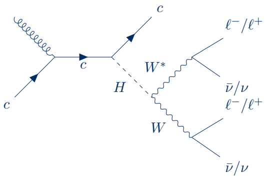
$H + c$ Final State

- Analysis goal: constrain **c Yukawa coupling**
- Look for $H + c$ events in $\mathbf{H(WW \to 2\ell 2\nu)}$ channel: H couples to c



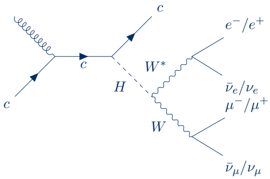
- Focus on events with **1 electron & 1 muon**

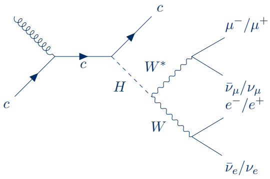
Challenges

- Large backgrounds
- Dirty event signature (MET)

Advantages

- High Branching Ratio
- Different dominant uncertainties
- Rich event topology, useful for S vs B MVA training





Possibly explore: **dimuon** and **dielectron** final states (possibly insensitive)

Run 2 Analysis Strategy (AN-23-102)

Credit: Aachen and Brown group

- Standard WW selection
- Tag a c-jet
- Due to different background compositions, Signal Region (SR) split in 2:
  1. $N_{c\text{-jet}} = 1$
  2. $N_{c\text{-jet}} > 1$

  $\rightarrow$ Increases sensitivity (8 % for Run 2)
- Define top Control Region (CR) enriched in tt events to derive top normalization factor
- Define a high transverse mass CR to build a CR where both $H + c$ and Higgs background contributions are negligible
- Use MVA (Boosted Decision Tree) to optimise signal recognition and separate it from other H production processes
- Modelling of background, main tt background in particular, with simultaneous fit in CRs and SRs

Run 2 Results

- Analysis finished (data unblinded) and paper submitted
- Full Run 2 dataset, integrated luminosity: 138 fb^{-1}
- Measured $H + c$ signal strength modifier (μ_{cH}) when:
  - H background fixed to SM (1 POI):

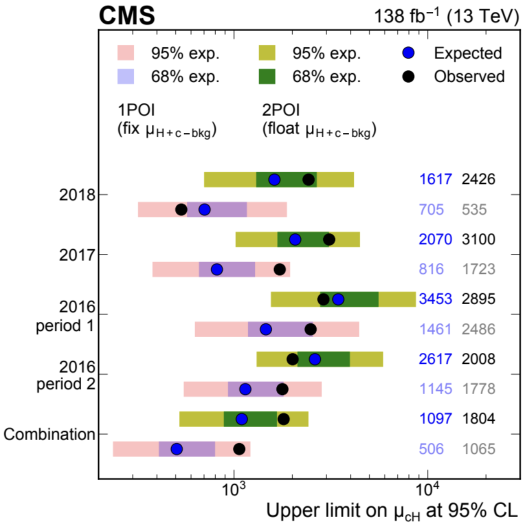
$$390^{+380}_{-270}$$

  - H bkg is fit parameter (2 POI):

$$230^{+1000}_{-640}$$

- Observed (expected) upper limit on μ_{cH} at 95 % CL, for 1 POI, is:

$$1065\ (506)$$



Run 3 Analysis Strategy (WIP)

- Same basic HWW selection as **AN-23-102** (Run 2 analysis)
- $Z \to \ell\ell$ CR is already validated
- Generate signal MC sample
- Due to different background compositions, Signal Region (SR) split in 2:
  1. $N_{c\text{-jet}} = 1$
  2. $N_{c\text{-jet}} > 1$

  $\rightarrow$ Increases sensitivity (8 % for Run 2)
- Define top CR enriched in tt events to derive top normalization factor
- Define a high mass CR to build a CR where both $H + c$ and Higgs background contributions are negligible
- Use MVA (Neural Network) to optimise signal recognition and separate it from other H production processes. Input features inspired by Run 2 analysis
- Modelling of the background, in particular the main tt background, with a simultaneous fit in the CRs and SRs using MVA classification

Samples

Currently included: full 2022 and 2023 datasets

Data Samples

- **Muon**: 2022CDEFG, 2023CD from 22Sep2023
  (incl. SingleMuon and DoubleMuon for 2022C)
- **EGamma**: 2022CDEFG, 2023CD from 22Sep2023
- **MuonEG**: 2022CDEFG, 2023CD from 22Sep2023

Non-Higgs MC Samples

- **tt**
  (powheg, pythia8)
- **Single Top**
  (powheg, pythia8)
- **W + Jets**
  (amcatnlo FXFX merging, pythia8)
- **ZZ, WZ, WW**
  (amcatnlo FXFX merging, pythia8)
- **WG**, binned in photon p_T
  (amcatnlo FXFX merging, pythia8)
- **DY + Jets**
  (amcatnlo FXFX merging, pythia8)

Private MC Samples

- **H+c**
  (amcatnlo FXFX merging, pythia8)
- **H+b**
  (amcatnlo FXFX merging, pythia8)

Higgs MC Samples

- **ggH, VBF**
  - **H $\rightarrow$ WW**
    (powheg-jhugen752, pythia8)
  - **H $\rightarrow$ $\tau\tau$**
    (amcatnlo FXFX merging, pythia8)
- **ZH**
  - **(qq)ZH**
    (powheg-minlo-HZJ-jhugenv752, pythia8)
  - **ggZH**
    (powheg-jhugenv752, pythia8)
- **WH**
  - **H $\rightarrow$ WW**
    (powheg-minlo-jhugen, pythia8)
  - **H $\rightarrow$ $\tau\tau$**
    (powheg-minnlo, pythia8)
- **ttH**
  - **H $\rightarrow$ bb/cc/Zγ**
    (powheg, pythia8)
  - **H $\rightarrow$ $\gamma\gamma$**
    (amcatnlo FXFX merging, madspin, pythia8)

Very loose object selections - Base

The basic selection on which the CRs and SRs are based.

Object	Criteria	Selection
Muon	p_T	$> 10\,\text{GeV}$
	$\lvert\eta\rvert$	< 2.4
	ID	Tight
	Iso	Tight
Electron	p_T	$> 10\,\text{GeV}$
	$\lvert\eta\rvert$	< 2.5
	ID	mvaIso_WP80
	$\Delta R(\mathrm{e}, \mu)$	> 0.4
Jet	p_T	$> 20\,\text{GeV}$
	$\lvert\eta\rvert$	< 2.5
	jetId	$= 6$ (tightLepVeto)
	$\Delta R(\mathrm{J}, \mathrm{e}/\mu)$	> 0.4
c-jet	is a Jet	
		ParticleNet b-tagger medium wp
		2022 preEE: $CvB > 0.306$ AND $CvL > 0.160$
	c-tagged	2022 postEE: $CvB > 0.304$ AND $CvL > 0.160$
		2023 preBPix: $CvB > 0.353$ AND $CvL > 0.148$
		2023 postBPix: $CvB > 0.157$ AND $CvL > 0.109$
Lepton Pair	$p_T(\ell_1)$	$> 20\,\text{GeV}$
	$p_T(\ell_2)$	$> 10\,\text{GeV}$
	$q(\ell_1) + q(\ell_2)$	$= 0$
	$m_{\ell\ell}$	$> 12\,\text{GeV}$
	$p_T^{\ell\ell}$	$> 30\,\text{GeV}$

Object selection criteria

Criteria	Selection
Trigger	SingleMu
	SingleEle
	MuEle
MET filters	see below
nPV	> 0
$N(\ell\ell$ pair)	$= 1$
	$\ell\ell$ pair with
one e, one μ	lepton 1 is e and
	lepton 2 is μ or vice versa
p_T^{miss} (Puppi MET)	$> 45\,\text{GeV}$

Event selection criteria

MET Filters (POG Recommendation)
`goodVertices`
`globalSuperTightHalo2016Filter`
`EcalDeadCellTriggerPrimitiveFilter`
`BadPFMuonFilter`
`BadPFMuonDzFilter`
`hfNoisyHitsFilter`
`eeBadScFilter`
`ecalBadCalibFilter`

Triggers

To **avoid duplicate events** from different primary datasets, events are selected as follows:

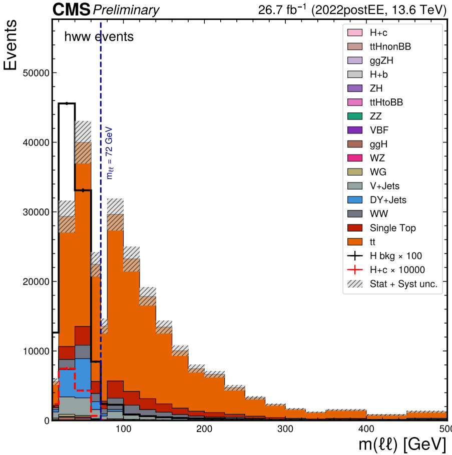
- from **EGamma** if event **passes SingleEle**
- from **Muon** if event **passes SingleMu**, while **failing SingleEle**
- from **MuonEG** if event **passes MuEle** triggers, while **failing SingleEle and SingleMu**

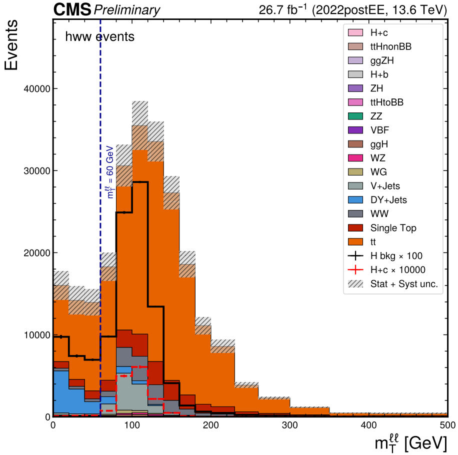
Flag	**HLT Paths**
SingleMu	`IsoMu24`
SingleEle	`Ele30_WPTight_Gsf`
MuEle	`Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL` `Mu12_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ`

HLT trigger paths and their corresponding defined flags

Validation of CR Definition

Illustration of choice of cuts used for CR definition





$$m_{\mathrm{T}}^{\ell\ell} = \sqrt{2 \cdot p_T^{\ell\ell} \cdot p_T^{\mathrm{miss}} \cdot (1 - \cos[\Delta\phi(\ell\ell,\ \mathrm{MET})])} \qquad (1)$$

CRs Selection

- Selections applied on top of Base selections.
- Distinction between the two CRs is shown in red.

Top CR

Criteria	**Selection**
$N(c\text{-jet})$	≥ 2
$m_{\ell\ell}$	$> 72\,\text{GeV}$
$m_T^{\ell\ell}$	$\leq 60\,\text{GeV}$
$m_T^{\ell_2}$	$> 30\,\text{GeV}$

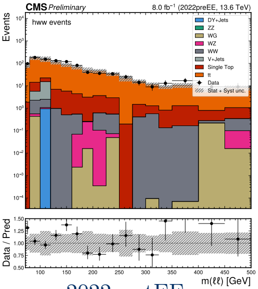
High Mass CR

Criteria	**Selection**
$N(\text{c-jet})$	≥ 1
$m_{\ell\ell}$	$> 72\,\text{GeV}$
$m_T^{\ell\ell}$	$> 60\,\text{GeV}$
$m_T^{\ell_2}$	$> 30\,\text{GeV}$

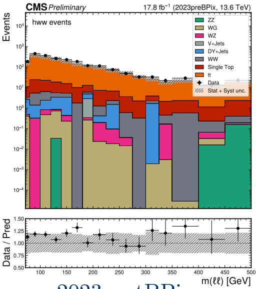
$$m_{\mathrm{T}}^{\ell 2} = \sqrt{2 \cdot p_T^{\ell 2} \cdot p_T^{\mathrm{miss}} \cdot (1 - \cos\left[\Delta\phi(\ell_2,\ \mathrm{MET})\right])} \tag{2}$$

Top Control Region: Dilepton Mass

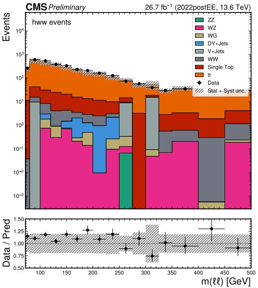
2022preEE



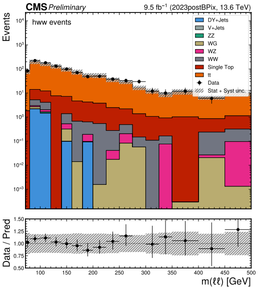
2023preBPix



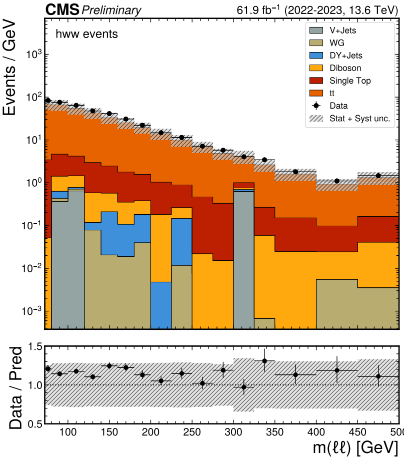
2022postEE



2023postBPix

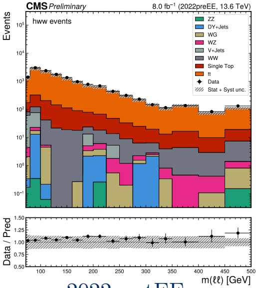


Full 2022-2023

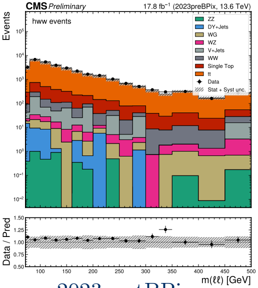


High Mass Control Region: Dilepton Mass

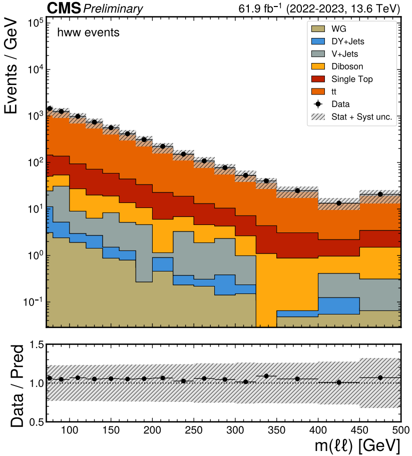
2022preEE



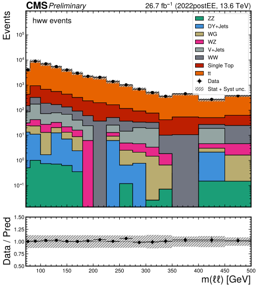
2023preBPix



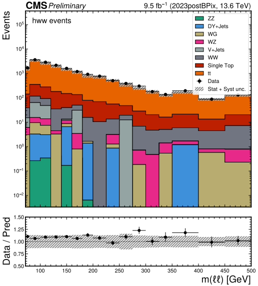
Full 2022-2023



2022postEE



2023postBPix



MVA Training Input Features Based on Run 2

- Run 2: Boosted Decision Tree trained in SR for
  - non-Higgs bkg vs signal
  - Higgs bkg vs signal
- Run 3: Neural Network with several categories:
  - Signal, tt, single top, diboson ...
  - Events not solely in SR: Base selection + ≥ 1 tagged c-jet $\rightarrow$ improves statistics by 15 %

Lepton kinematics	**(Transverse) mass**	$\Delta\phi$	ΔR
$p_T^{\ell\ell}$	$m_T^{\ell_1}$	$\Delta\phi(\ell\ell + \mathrm{MET}, \mathrm{c})$	$\Delta R(\ell\ell, \ell_1)$
$p_T^{\ell_1}$	$m_T^{\ell_2}$	$\Delta\phi(\ell_1, \mathrm{MET})$	$\Delta R(\ell\ell, \ell_2)$
$p_T^{\ell_2}$	$m_{\ell\ell}$	$\Delta\phi(\ell_2, \mathrm{MET})$	$\Delta R(\ell\ell, \mathrm{c})$

Charm tagging	**Jet kinematics**	**Vertex**	**MET kinematics**
c-jet ParticleNet CvL	p_T^c	nSV	p_T^{miss}
c-jet ParticleNet CvB			

α-weighted Feature Importance

Investigate: **which Run 3 features most important** to distinguish Signal from Background.

Based on integrated gradients approach to calculate feature similarity:
checks difference between gradients of a feature for signal vs gradients for a particular background j.

Formula:

$$I_j = \alpha_j \cdot \sum_f \left| \frac{\partial (P_{\mathrm{H+c}} - P_j)}{\partial x_f} \right| \mathrm{std}\,(x_f)$$

where:

$$\alpha_j = \sigma\left(\frac{\cos(\mathbf{w}_{\mathrm{H+c}}, \mathbf{w}_j)}{\tau} \right)$$

See arXiv:1703.01365.

All Tested Input Features

Lepton kinematics

$p_{\mathrm{T}}^{\ell\ell}$
$p_{\mathrm{T}}^{\ell_1}$
$p_{\mathrm{T}}^{\ell_2}$

Charm tagging

c-jet ParticleNet CvL
c-jet ParticleNet CvB
jet 1 ParticleNet CvL
jet 1 ParticleNet CvB
jet 2 ParticleNet CvL
jet 2 ParticleNet CvB
jet 3 ParticleNet CvL
jet 3 ParticleNet CvB

(Transverse) mass

$m_{\mathrm{T}}^{\ell_1}$
$m_{\mathrm{T}}^{\ell_2}$
$m_{\ell\ell}$
$\boldsymbol{m_{\mathrm{T}}^{\ell\ell}}$

Jet kinematics

p_{T}^{c}
$\boldsymbol{p_{\mathrm{T}}^{\mathrm{jets}}}$
$\boldsymbol{\eta(\mathrm{jets})}$
$\boldsymbol{\phi(\mathrm{jets})}$
$\boldsymbol{p_{\mathrm{T}}^{j_1}}$
$\boldsymbol{p_{\mathrm{T}}^{j_2}}$
$\boldsymbol{p_{\mathrm{T}}^{j_3}}$

$\Delta\phi$

$\Delta\phi(\ell_1, \mathrm{MET})$
$\Delta\phi(\ell_2, \mathrm{MET})$
$\Delta\phi(\ell\ell + \mathrm{MET}, \mathrm{c})$
$\boldsymbol{\Delta\phi(\ell\ell, \mathrm{MET})}$
$\boldsymbol{\Delta\phi(\ell_1 + \mathrm{MET}, \mathrm{c})}$
$\boldsymbol{\Delta\phi(\ell_1 + \mathrm{MET}, j_1)}$
$\boldsymbol{\Delta\phi(\ell_1 + \mathrm{MET}, j_2)}$
$\boldsymbol{\Delta\phi(\ell_1 + \mathrm{MET}, j_3)}$
$\boldsymbol{\Delta\phi(\ell\ell + \mathrm{MET}, j_1)}$
$\boldsymbol{\Delta\phi(\ell\ell + \mathrm{MET}, j_2)}$
$\boldsymbol{\Delta\phi(\ell\ell + \mathrm{MET}, j_3)}$

Vertex

nSV

ΔR

$\Delta R(\ell\ell, \ell_1)$
$\Delta R(\ell\ell, \ell_2)$
$\Delta R(\ell\ell, \mathrm{c})$
$\boldsymbol{\Delta R(\ell\ell, j_1)}$
$\boldsymbol{\Delta R(\ell\ell, j_2)}$
$\boldsymbol{\Delta R(\ell\ell, j_3)}$

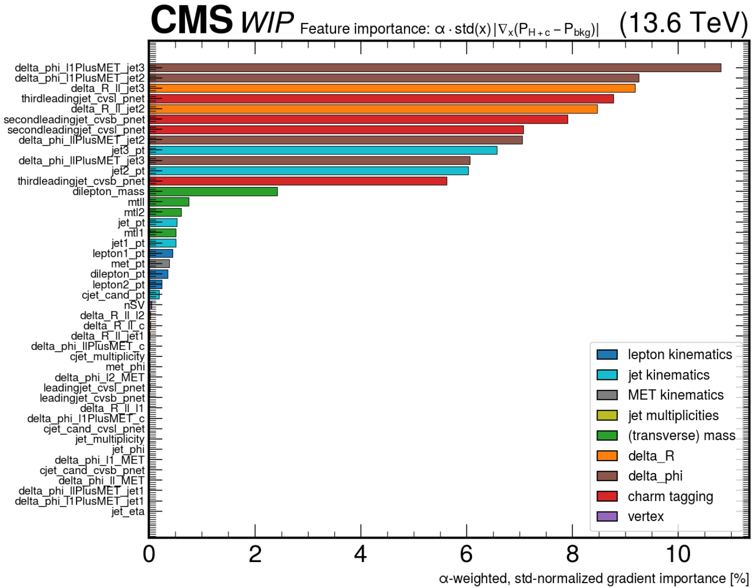
MET kinematics

$p_{\mathrm{T}}^{\mathrm{miss}}$
$\boldsymbol{\phi(\mathrm{MET})}$

Jet multiplicities

$\boldsymbol{N(\mathrm{jets})}$
$\boldsymbol{N(\mathrm{c\text{-}jets})}$

Feature Ranking



New important features w.r.t. Run 2 mostly related to 2nd and 3rd leading jets:

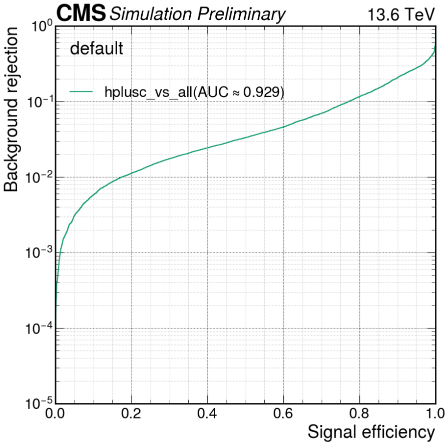
- $\Delta\phi(\ell_1 + \mathrm{MET},\ \mathrm{jet}\ 2/3)$
- $\Delta\phi(\ell\ell + \mathrm{MET},\ \mathrm{jet}\ 2/3)$
- $\Delta R(\ell\ell\ \mathrm{jet}\ 2/3)$
- $p_T(\mathrm{jet}\ 2/3)$
- $CvsL(\mathrm{jet}\ 2/3)$
- $CvsB(\mathrm{jet}\ 2/3)$

→ Highlights possibility of removing any features relying on a c-tagged jet.

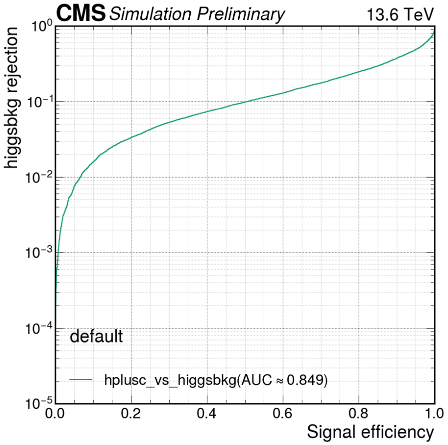
MVA Performance

Using basic MVA model: already decent performance.

Signal vs all bkg: AUC = 0.929



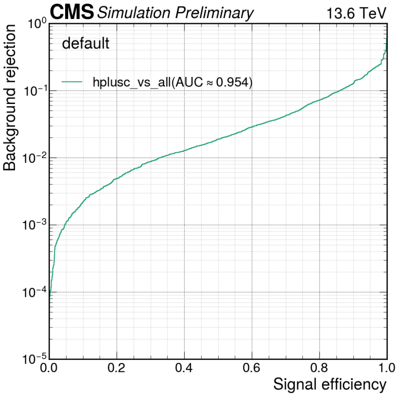
Signal vs H bkg: AUC = 0.849



MVA Performance

With a more advanced model, we can improve this bit by bit.
Below: result for only using the 17 feautures based on Run 2.

Signal vs all bkg: AUC = 0.954



More details on studies for obtaining a better model: Chirayu's talk tomorrow.

Systematic Uncertainties: current inventory

25 shape · **9 lnN** · `rate_tt` free rateParam · autoMCStats (threshold 10) in all 6 channels

Shape — event weight

`pileup`	pileup profile
`ps_isr`, `ps_fsr`	parton shower
`scalevar_muR`, `_muF`	renorm./fact. scale
`scalevar_muR_muF`	correlated variation
`lhe_pdf`	NNPDF replicas
`lhe_alphaS`	α_S variation
`muon_id`, `muon_iso`	muon efficiency
`electron_id`	electron efficiency
`electron_reco_RecoBelow20/20to75/RecoAbove75`	reco, p_T-binned (<20, 20–75, >75 GeV)
`CMS_ctag2d_2022`	2D CvL/CvB SF
`CMS_negrw_vjets`	negative-weight rw.
`top_pt`	$t\bar{t}$ top-p_T rw.
`higgs_HF`	Higgs heavy-flavour

Shape — object shift

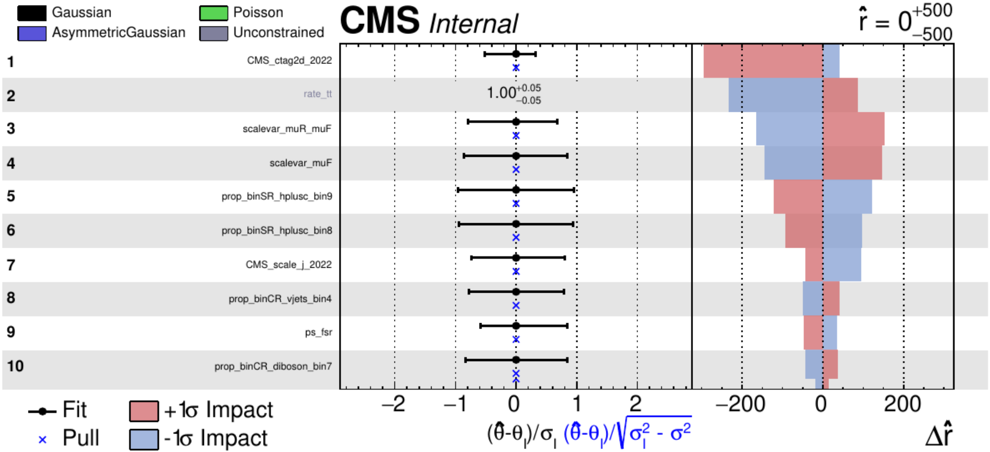
`CMS_scale_j_2022`	JES
`CMS_res_j_2022`	JER
`CMS_scale_e_2022`	electron scale
`CMS_res_e_2022`	electron res.
`CMS_scale_m_2022`	muon scale
`CMS_res_m_2022`	muon res.
`MET_unclustered`	PuppiMET unclust.

lnN

`lumi_13p6TeV`	1.4%
`xsec_st`	$-1.3/+1.7\%$
`xsec_diboson`	3.7%
`xsec_vjets`	2.7%
`xsec_higgsbkg`	5.0%
`flavor_composition_ggH`	6.6%
`BR_HtoWW`, `BR_Htautau`	1.0%
`xsec_hplusc_4FS_5FS`	30%

$t\bar{t}$ carries no `xsec` term — its normalisation floats freely via `rate_tt`, pinned by the 94%-pure tt control region.

Nuisance impacts — top 10

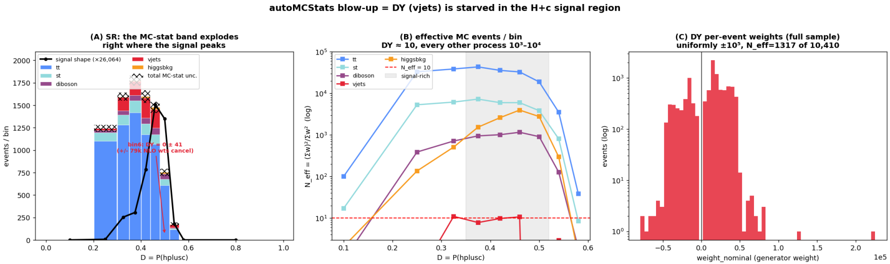


- The header $\hat{r} = 0\,^{+500}_{-500}$ is a `plotImpacts` rounding artefact — the fit returns $\hat{r} = 0.99996,\ ^{+483}_{-513}$, exactly the injected Asimov value

Negative-weight reweighting — the problem

aMC@NLO events carry **signed** weights, so the yield estimator $\hat{N}_B = \sum_{i \in B} w_i$ is a **difference of two large numbers**. The mean is unbiased, but the variance scales with $\sum |w_i|^2$ — the *unsigned* statistics — so the effective sample size $N_{eff} = (\sum w)^2 / \sum w^2$ collapses.

16.4% of our V+jets events carried a negative weight.



(A) the MC-stat band explodes where the signal peaks — one bin was DY $= 0 \pm 41$ from $\pm$79k weights cancelling · **(B)** N_{eff} per bin: V+jets $\approx$10, every other process 10^3–10^4

Negative-weight reweighting — the method

Write the signed generator density as a difference of two *positive* densities, $\mathrm{PDF} = a\,\mathrm{PDF}_+ - b\,\mathrm{PDF}_-$ with $a - b = 1$. Defining $P_+(\vec{x}) = a\mathrm{PDF}_+/(a\mathrm{PDF}_+ + b\mathrm{PDF}_-)$:

$$\mathrm{PDF}(\vec{x}) = \underbrace{\big(2P_+(\vec{x}) - 1\big)}_{g(\vec{x})} \cdot \underbrace{\big(a\mathrm{PDF}_+ + b\mathrm{PDF}_-\big)}_{\text{unsigned density}}$$

A classifier learns P_+ from generator-level kinematics; each event is then reweighted by $g(\vec{x})$, so the estimator $\sum |w|\, g$ samples the **unsigned** density instead of differencing large signed sums.

	V+jets (SR)
N_{eff}	280 → **1170**
statistical error	5.98% → **2.92%**
rate	1508 → 1523

- Training-region closure **0.994** on 9.8M events
- Renormalised **per dataset** — yield restored exactly, variance reduction kept

The rate barely moves while the statistical error halves — the signature of a statistics gain, not a normalisation change.

Method follows **Palmer & Kronheim**, arXiv:2510.16217.

Comparison with the Run 2 analysis

	Expected UL
AN-23-102 scaled to 26.7 fb^{-1}	**1144**
this analysis	**1034**
our statistical only	**641**

- **Better than** the published analysis scaled to our luminosity — a **9.6% improvement** on **5× less data**

Conclusion

- 2022 and 2023 are both included now
- $Z \to \ell\ell$ control region has been validated (in common with $H \to ZZ$ analysis)
- MC signal and $H + b$ background samples generated. Central Production underway
- Investigating best input features and architecture for MVA
- Preliminary estimates show that Run 3 analysis already promises to be competitive with Run 2, but we need to stress that this is still very much WIP

Next steps:

- Converge on lepton working points used in the main HWW analysis
- Optimize MVA training using calibrated PNet features
- Study systematic uncertainty when using 3 leading jets instead of 1 tagged c jet
- Extend to 2024 data (Run 3, NanoAODv15) once all MC is available

Backup

2022 Experimental Data Samples

- Below, experimental data samples used from 2022preEE and 2022postEE campaigns

Campaign	DAS Path	Luminosity [1/fb]

2022preEE	`Muon/Run2022C-22Sep2023-v1/NANOAOD`
	`DoubleMuon/Run2022C-22Sep2023-v1/NANOAOD`
	`SingleMuon/Run2022C-22Sep2023-v1/NANOAOD`
	`EGamma/Run2022C-22Sep2023-v1/NANOAOD`
	`MuonEG/Run2022C-22Sep2023-v1/NANOAOD`
		5.0104

	`Muon/Run2022D-22Sep2023-v1/NANOAOD`
	`EGamma/Run2022D-22Sep2023-v1/NANOAOD`
	`MuonEG/Run2022D-22Sep2023-v1/NANOAOD`
		2.9700

2022postEE	`Muon/Run2022E-22Sep2023-v1/NANOAOD`
	`EGamma/Run2022E-22Sep2023-v1/NANOAOD`
	`MuonEG/Run2022E-22Sep2023-v1/NANOAOD`
		5.8070

	`Muon/Run2022F-22Sep2023-v2/NANOAOD`
	`EGamma/Run2022F-22Sep2023-v1/NANOAOD`
	`MuonEG/Run2022F-22Sep2023-v1/NANOAOD`
		17.7819

	`Muon/Run2022G-22Sep2023-v1/NANOAOD`
	`EGamma/Run2022G-22Sep2023-v2/NANOAOD`
	`MuonEG/Run2022G-22Sep2023-v1/NANOAOD`
		3.0828

2023preBPix Experimental Data Samples

- Below, experimental data samples used from 2023preBPix campaign

Campaign	**DAS Path**	**Luminosity** [1/fb]
2023preBPix	MuonEG/Run2023C-22Sep2023_v1-v1/NANOAOD	17.7940
	MuonEG/Run2023C-22Sep2023_v2-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v3-v1/NANOAOD	
	MuonEG/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	EGamma1/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon0/Run2023C-22Sep2023_v4-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v1-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v2-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v3-v1/NANOAOD	
	Muon1/Run2023C-22Sep2023_v4-v2/NANOAOD	

2023postBPix Experimental Data Samples

- Below, experimental data samples used from 2023postBPix campaign

Campaign	**DAS Path**	**Luminosity** [1/fb]
2023postBPix	`MuonEG/Run2023D-22Sep2023_v1-v1/NANOAOD`	9.4500
	`MuonEG/Run2023D-22Sep2023_v2-v1/NANOAOD`	
	`EGamma0/Run2023D-22Sep2023_v1-v1/NANOAOD`	
	`EGamma0/Run2023D-22Sep2023_v2-v1/NANOAOD`	
	`EGamma1/Run2023D-22Sep2023_v1-v1/NANOAOD`	
	`EGamma1/Run2023D-22Sep2023_v2-v1/NANOAOD`	
	`Muon0/Run2023D-22Sep2023_v1-v1/NANOAOD`	
	`Muon0/Run2023D-22Sep2023_v2-v1/NANOAOD`	
	`Muon1/Run2023D-22Sep2023_v1-v1/NANOAOD`	
	`Muon1/Run2023D-22Sep2023_v2-v1/NANOAOD`	

Background MC Samples 2022 and 2023

Process	DAS Path	$\sigma \times$ BR [pb]	Source
tt			
$tt \to 2W + 2b \to 2\ell 2\nu + 2b$	`TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8`	98.578	Twiki for NNLO σ, arxiv:2201.07861 for BR
$tt \to 2W + 2b \to (\ell\nu + 2q) + 2b$	`TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8`	406.263	Twiki for NNLO σ, arxiv:2201.07861 for BR
$tt \to 2W + 2b \to 4q + 2b$	`TTto4Q_TuneCP5_13p6TeV_powheg-pythia8`	418.574	Twiki for NNLO σ, arxiv:2201.07861 for BR
Single Top			
$t + W^- \to (W^+ + b) + W^- \to 2q + b + 2q$	`TWminusto4Q_TuneCP5_13p6TeV_powheg-pythia8`	19.918	Twiki for NNLO σ, arxiv:2201.07861 for BR
$t + W^- \to (W^+ + b) + W^- \to \ell\nu + b + 2q$	`TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8`	19.332	Twiki for NNLO σ, arxiv:2201.07861 for BR
$t + W^- \to (W^+ + b) + W^- \to \ell\nu + b + \ell\nu$	`TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8`	4.691	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + W^+ \to (W^- + \bar{b}) + W^+ \to 2q + \bar{b} + 2q$	`TbarWplusto4Q_TuneCP5_13p6TeV_powheg-pythia8`	19.918	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + W^+ \to (W^- + \bar{b}) + W^+ \to \ell\nu + \bar{b} + 2q$	`TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8`	19.332	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + W^+ \to (W^- + \bar{b}) + W^+ \to \ell\nu + \bar{b} + \ell\nu$	`TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8`	4.691	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + q \to (W^- + \bar{b}) + q \to 2q + \bar{b} + q$	`TbarQto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8`	58.703	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + q \to (W^- + \bar{b}) + q \to \ell\nu + \bar{b} + q$	`TbarQtoLNu-t-channel_TuneCP5_13p6TeV_powheg-pythia8`	28.488	Twiki for NNLO σ, arxiv:2201.07861 for BR
$t + \bar{q} \to (W^+ + b) + \bar{q} \to 2q + b + \bar{q}$	`TQbarto2Q-t-channel_TuneCP5_13p6TeV_powheg-pythia8`	97.614	Twiki for NNLO σ, arxiv:2201.07861 for BR
$t + \bar{q} \to (W^+ + b) + \bar{q} \to \ell\nu + b + \bar{q}$	`TQbartoLNu-t-channel_TuneCP5_13p6TeV_powheg-pythia8`	47.372	Twiki for NNLO σ, arxiv:2201.07861 for BR
$\bar{t} + b \to (W^- + \bar{b}) + b \to \ell\nu + \bar{b} + b$	`TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8`	1.481	Twiki for NNLO σ, arxiv:2201.07861 for BR
$t + \bar{b} \to (W^+ + b) + \bar{b} \to \ell\nu + b + \bar{b}$	`TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8`	2.367	Twiki for NNLO σ, arxiv:2201.07861 for BR
V + Jets			
$W + 0,\ 1,\ 2$ jets $\to \ell\nu +$ jets	`WtoLNu-2Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	67710.000	XSDB
Diboson			
$ZZ \to ...$	`ZZ_TuneCP5_13p6TeV_powheg-pythia8`	16.70	CMS AN-24-078
$WZ \to ...$	`WZ_TuneCP5_13p6TeV_powheg-pythia8`	54.30	CMS AN-24-078
$WW \to ...$	`WW_TuneCP5_13p6TeV_powheg-pythia8`	122.30	CMS AN-24-078

Background MC samples and cross sections values for both 2022 and 2023

Background MC Samples 2022 and 2023

Process	DAS Path	$\sigma \times$ BR [pb]	Source
WG (Diboson)			
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1$ jets	`WGtoLNuG-1Jets_PTG-10to100_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	662.2	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1$ jets	`WGtoLNuG-1Jets_PTG-100to200_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	2.221	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1$ jets	`WGtoLNuG-1Jets_PTG-200to400_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	0.2908	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1$ jets	`WGtoLNuG-1Jets_PTG-400to600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	0.02231	XSDB
$W\gamma \rightarrow \ell\nu + \gamma + 0, 1$ jets	`WGtoLNuG-1Jets_PTG-600_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	0.004907	XSDB
DY + Jets			
$qq \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	`DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	6688.0	XSDB
$qq \rightarrow Z/\gamma^* + 0, 1, 2 \text{ jets} \rightarrow 2\ell + \text{jets}$	`DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8`	20950.0	XSDB
Higgs Production			
$gg \rightarrow H \rightarrow WW \rightarrow 2\ell 2\nu$	`GluGluHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8`	1.0825	YR4 SM for σ , arxiv:2201.07861 for BR
$qq + VV \rightarrow qq + H \rightarrow qq + WW \rightarrow qq + 2\ell 2\nu$	`VBFHto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugen752-pythia8`	0.09305	YR4 SM for σ , arxiv:2201.07861 for BR
$qq \rightarrow Z + H \rightarrow ... + WW \rightarrow ... + 2\ell 2\nu$	`ZH_ZtoAll_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-HZJ-jhugenv752-pythia8`	0.021559	YR4 SM for σ , arxiv:2201.07861 for BR
$gg \rightarrow Z + H \rightarrow ... + WW \rightarrow ... + 2\ell 2\nu$	`GluGluZH_ZtoAll_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-jhugenv752-pythia8`	0.003094	YR4 SM for σ , arxiv:2201.07861 for BR
$gg \rightarrow H \rightarrow 2\tau$	`GluGluHto2Tau_M-125_TuneCP5_13p6TeV_amcatnloFXFX-pythia`	2.976691	YR4 SM for σ , arxiv:2201.07861 for BR
$qq + VV \rightarrow qq + H \rightarrow 2\tau$	`VBFHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-pythia8`	0.255873	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow \ell\nu + WW \rightarrow \ell\nu + 2\ell 2\nu$	`WplusH_WtoLNu_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8`	0.006625	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow qq + WW \rightarrow qq + 2\ell 2\nu$	`WminusH_Wto2Q_Hto2Wto2L2Nu_M-125_TuneCP5_13p6TeV_powheg-minlo-jhugen-pythia8`	0.008719	YR4 SM for σ , arxiv:2201.07861 for BR
$W^- + H \rightarrow ... + 2\tau$	`WminusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8`	0.035615	YR4 SM for σ , arxiv:2201.07861 for BR
$W^+ + H \rightarrow ... + 2\tau$	`WplusHTo2TauUncorrelatedDecay_M-125_TuneCP5_13p6TeV_powheg-minnlo-pythia8`	0.055763	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow ... + 2c$	`ttHto2C_M-125_TuneCP5_13p6TeV_powheg-pythia8`	0.016509	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow ... + 2\gamma$	`ttHtoGG_M-125_TuneCP5_13p6TeV_amcatnloFXFX-madspin-pythia8`	0.001296	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow ... + (Z + \gamma) \rightarrow .. + ... + \gamma$	`ttHtoZG_ZtoAll_M-125_TuneCP5_13p6TeV_powheg-pythia8`	0.000875	YR4 SM for σ , arxiv:2201.07861 for BR
$tt + H \rightarrow ... + 2b$	`ttHto2B_M-125_TuneCP5_13p6TeV_powheg-pythia8`	0.332585	YR4 SM for σ , arxiv:2201.07861 for BR
$H + b \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.002214	
$H + c \rightarrow WW + b \rightarrow 2\ell 2\nu + b$	private sample	0.016972	

Background MC samples (+ private signal MC sample) and cross sections values for both 2022 and 2023

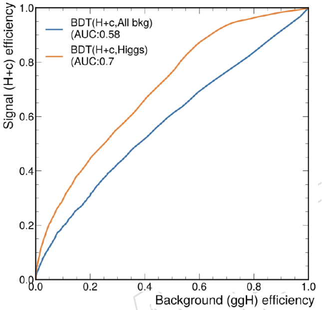
Missing Higgs MC Samples 2022 and 2023

Process	Sample(s)	Status / Remarks
$H + c$	`HPlusCharm`	In central production
$H + b$	`HPlusBottom`	In central production
$W + H \to \ell\nu + WW \to \ell\nu + 2\ell 2\nu$	`WplusH_WtoLNu_Hto2Wto2L2Nu`, `WminusH_WtoLNu_Hto2Wto2L2Nu`	WplusH available for 2022postEE, not for other years
$W + H \to qq + WW \to qq + 2\ell 2\nu$	`WplusH_Wto2Q_Hto2Wto2L2Nu`, `WminusH_Wto2Q_Hto2Wto2L2Nu`	WminusH available for 2022postEE, not for other years
$tt + H \to ... +$ non-bb	`ttHToNonbb`	Other non-bb samples, besides $2c$, 2γ, $Z\gamma$ needed?
$Z + H \to ... + 2\tau$	`ZHToTauTau` / `ZHTo2Tau`	Now available, but not yet used
$tt + H \to ... + 2\tau$	`ttHToTauTau` / `ttHTo2Tau`	Not available for all Run 3 years

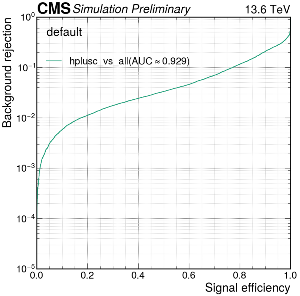
Missing MC samples for both 2022 and 2023

Run 2 vs Run 3 (v11) MVA

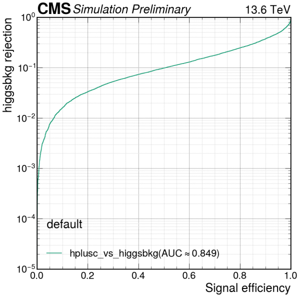
Run 2



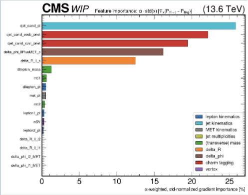
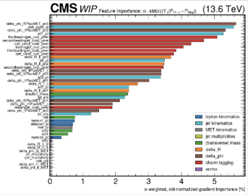
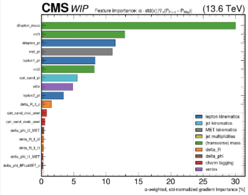
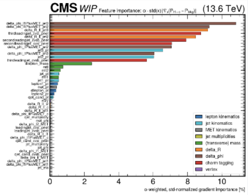
Run 3 Higgs vs all



Run 3 Higgs vs Higgs bkg



Input feature analysis for 2022preEE - normalized

	Run 2 features	All possible features
Base selection		
Base + at least 1 c-jet		

The selection of the c-jet is done with the medium working point from ParticleNet. Data is for 2022preEE.

Input feature analysis for 2022preEE

AUC overview
MVA model v6

Hplusc_vs_all	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.01

Hplusc_vs_diboson	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	0.993

Hplusc_vs_higgskg	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.02

Hplusc_vs_st	All possible / Run 2
Base selection	1.00
Base + at least 1 c-jet	0.997

Hplusc_vs_tt	All possible / Run 2
Base selection	1.01
Base + at least 1 c-jet	1.01

Hplusc_vs_vjets	All possible / Run 2
Base selection	1.00
Base + at least 1 c-jet	1.00

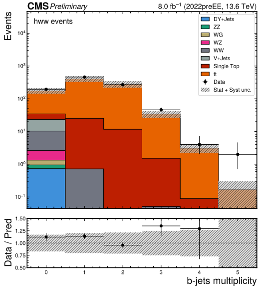
For some MVA categories the one selection is better, but for others, the other one is better. Generally the difference becomes larger in the worst performing categories and there the selection with the cjet cut does better.

Input feature analysis for 2022preEE

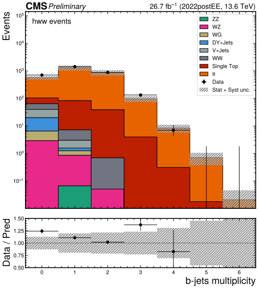
	Base selection	Base + at least 1 c-jet	Improvement by staying with Base
S/B	0.0043	0.0022	1.95
Significance S/sqrt(B)	7.29	2.48	2.94

Top Control Region: bjet_multiplicity

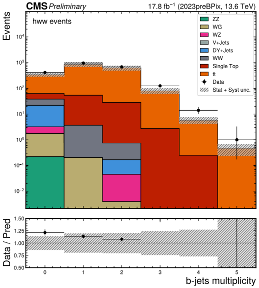
2022preEE



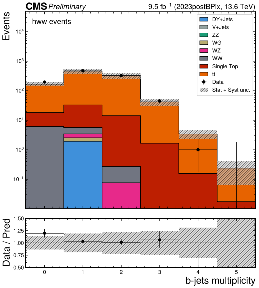
2022postEE



2023preBPix

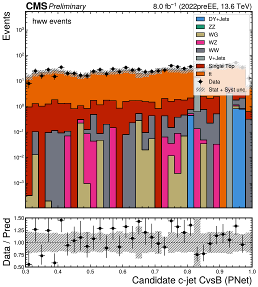


2023postBPix

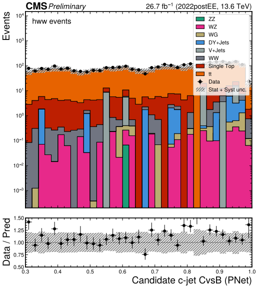


Top Control Region: `cjet_cand_cvsb_pnet`

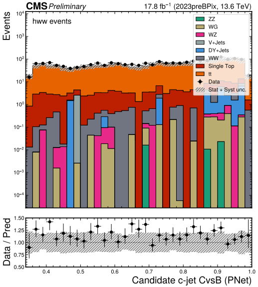
2022preEE



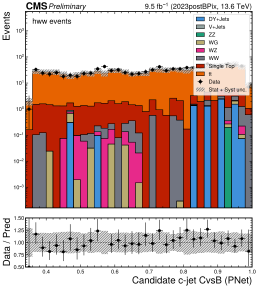
2022postEE



2023preBPix

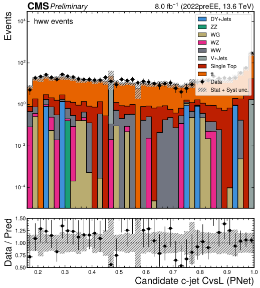


2023postBPix

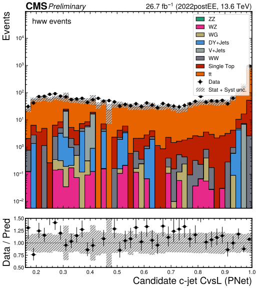


Top Control Region: `cjet_cand_cvsl_pnet`

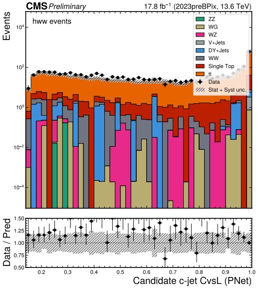
2022preEE



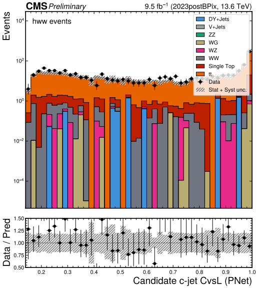
2022postEE



2023preBPix

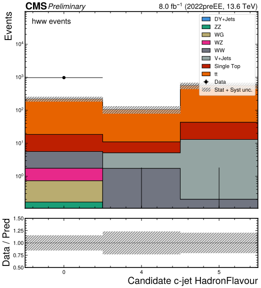


2023postBPix

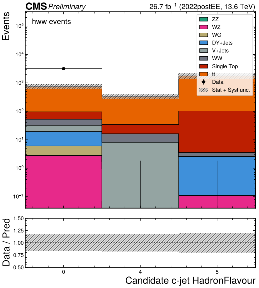


Top Control Region: `cjet_cand_flavour`

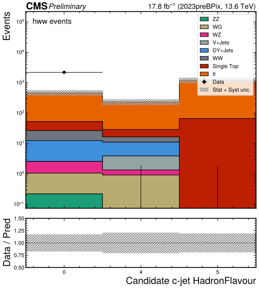
2022preEE



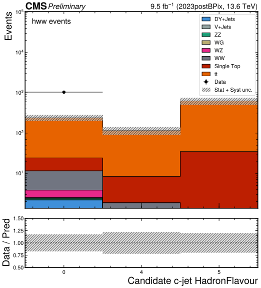
2022postEE



2023preBPix

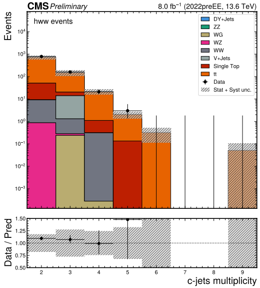


2023postBPix

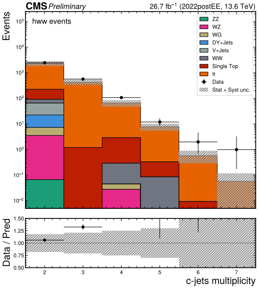


Top Control Region: `cjet_multiplicity`

2022preEE



2022postEE



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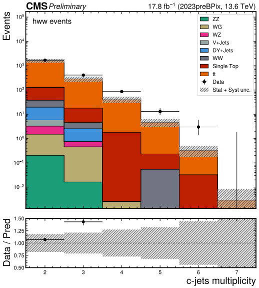

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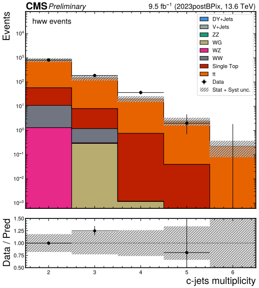
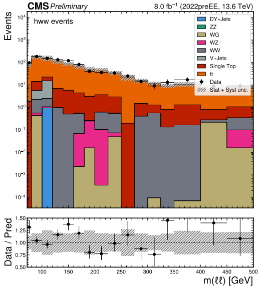
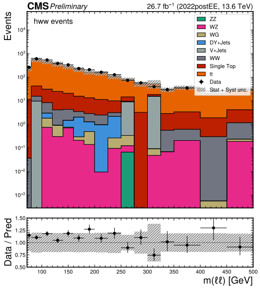


Top Control Region: dilepton_mass

2022preEE



2022postEE



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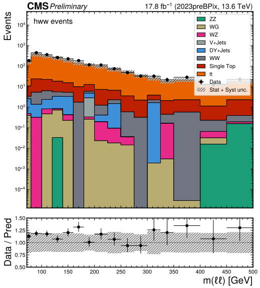

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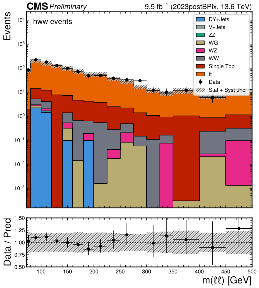
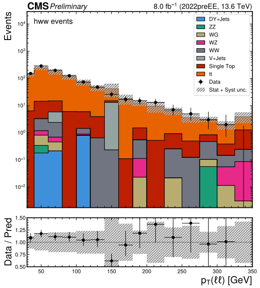
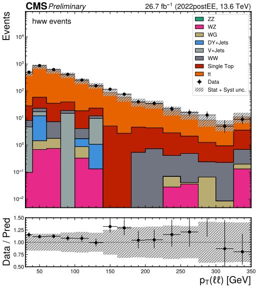


Top Control Region: dilepton_pt

2022preEE



2022postEE



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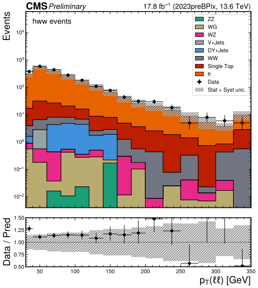

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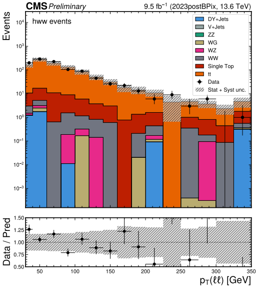
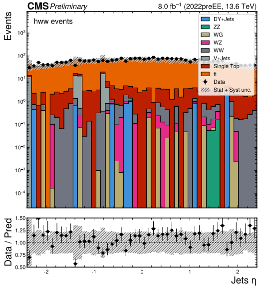
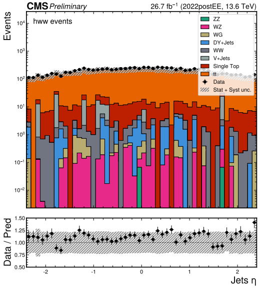


Top Control Region: `jet_eta`

2022preEE



2022postEE



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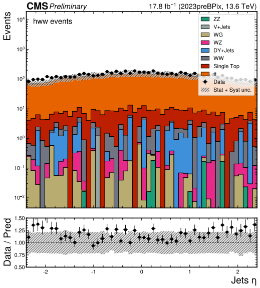

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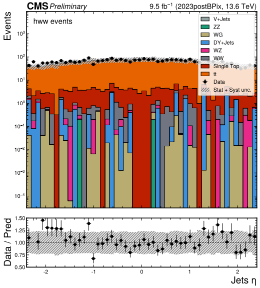
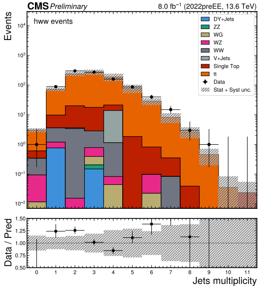
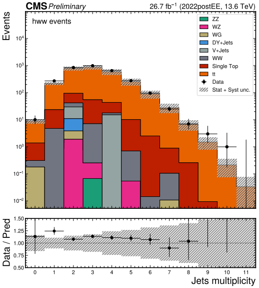


Top Control Region: jet_multiplicity

2022preEE



2022postEE



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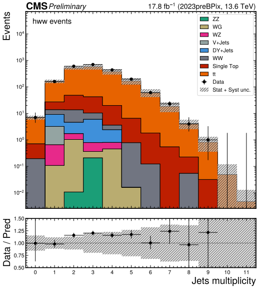

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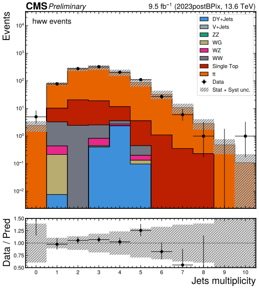
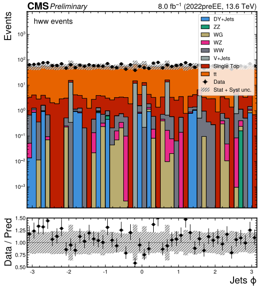
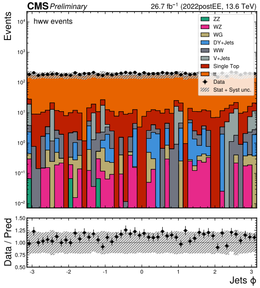


Top Control Region: `jet_phi`

2022preEE



2022postEE



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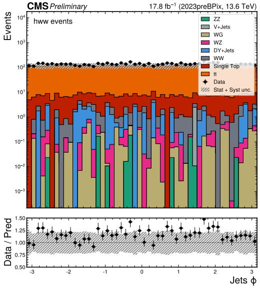

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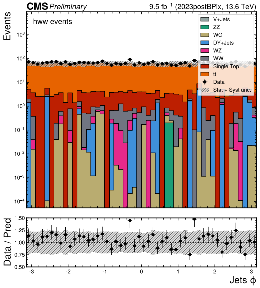
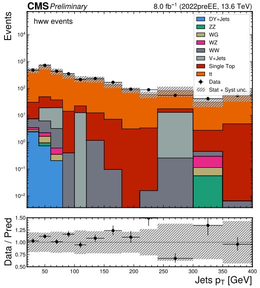
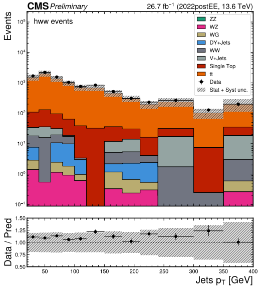


Top Control Region: jet_pt

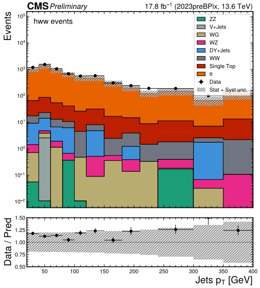
2022preEE



2022postEE



2023preBPix



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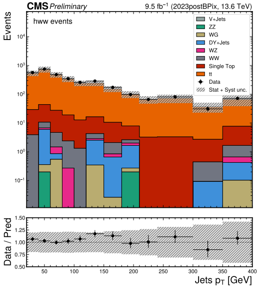
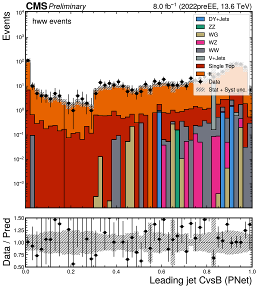
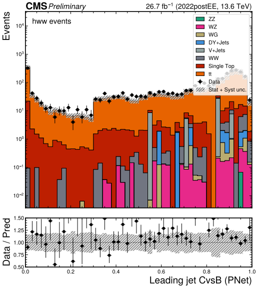
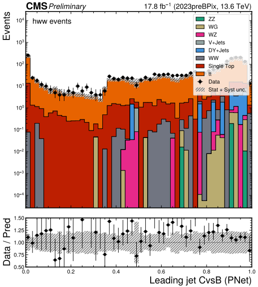


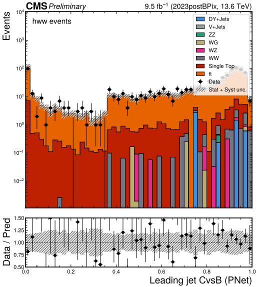
Top Control Region: `leadingjet_cvsb_pnet`

2022preEE	2022postEE	2023preBPix	2023postBPix



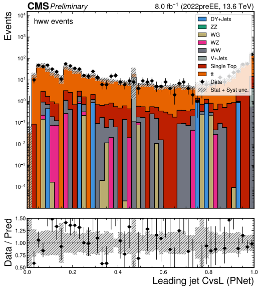




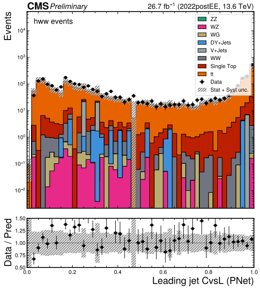


Top Control Region: `leadingjet_cvsl_pnet`

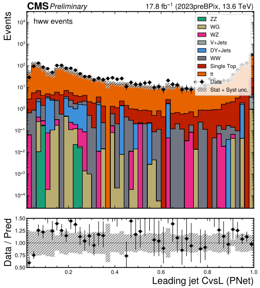
2022preEE



2022postEE



2023preBPix



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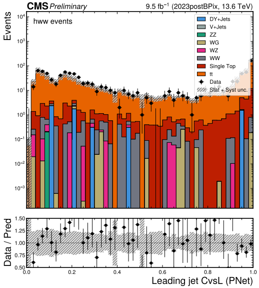
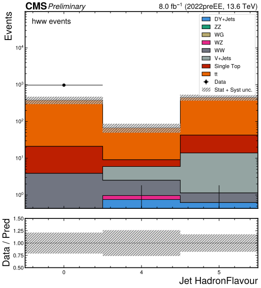
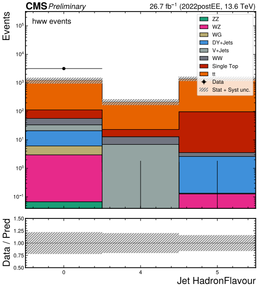


Top Control Region: `leadingjet_flavour`

2022preEE



2022postEE



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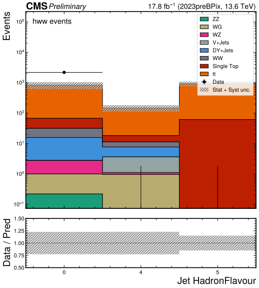
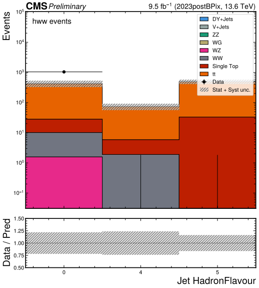
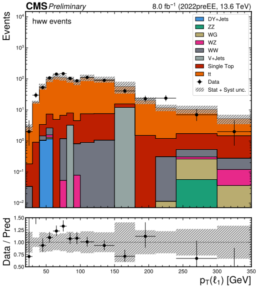


2023postBPix

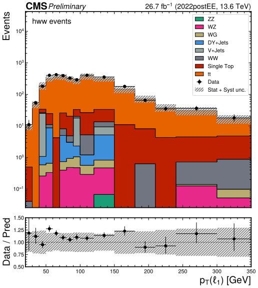


Top Control Region: `lepton1_pt`

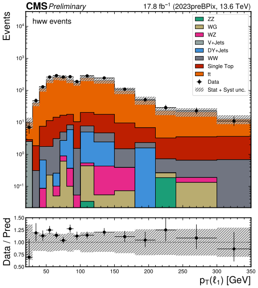
2022preEE



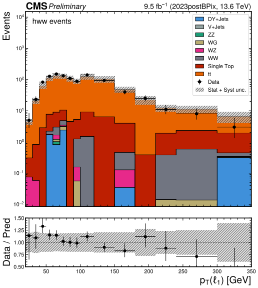
2022postEE



2023preBPix

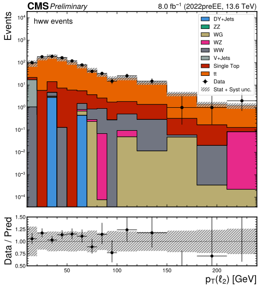


2023postBPix

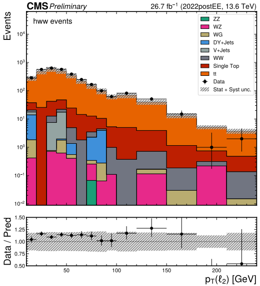


Top Control Region: `lepton2_pt`

2022preEE



2022postEE



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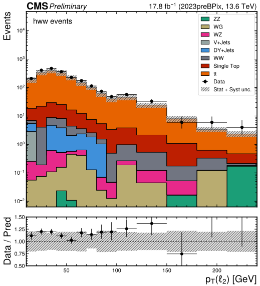

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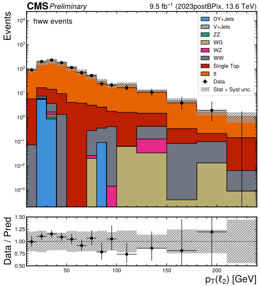
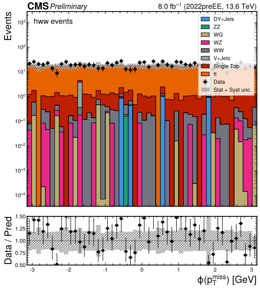
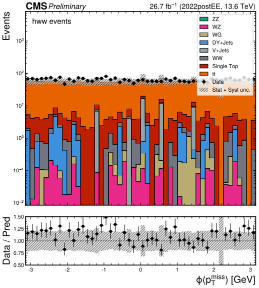


Top Control Region: met_phi

2022preEE



2022postEE



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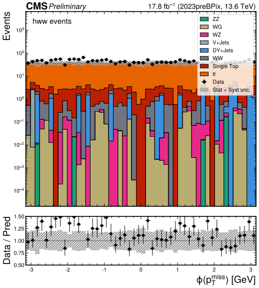
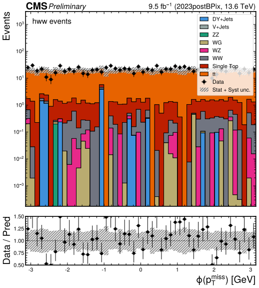
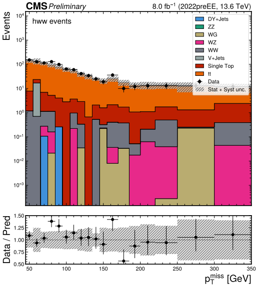


2023postBPix

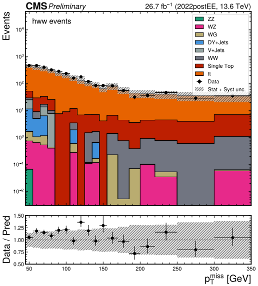


Top Control Region: `met_pt`

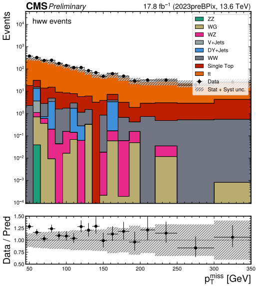
2022preEE



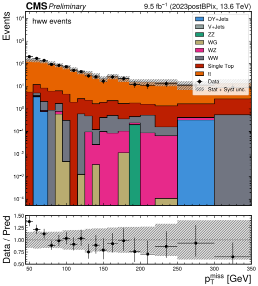
2022postEE



2023preBPix

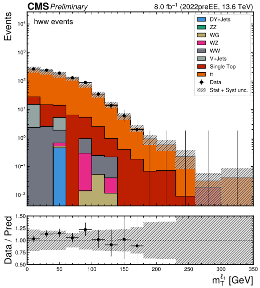


2023postBPix

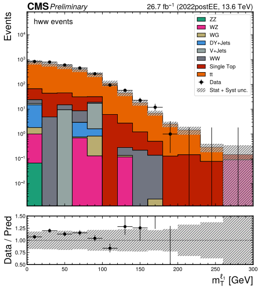


Top Control Region: `mtl1`

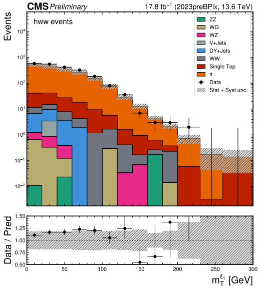
2022preEE



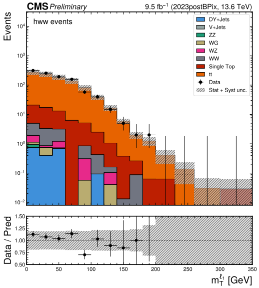
2022postEE



2023preBPix

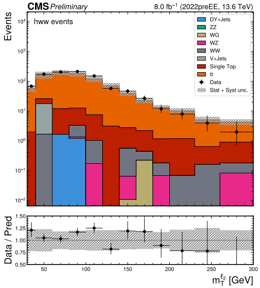


2023postBPix

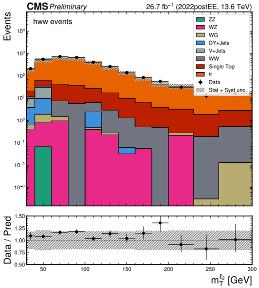


Top Control Region: `mtl2`

2022preEE



2022postEE



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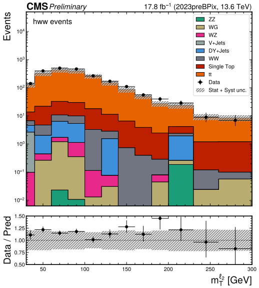
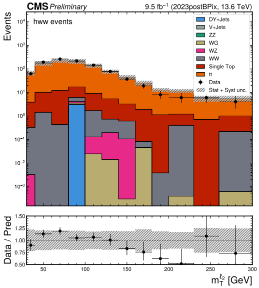
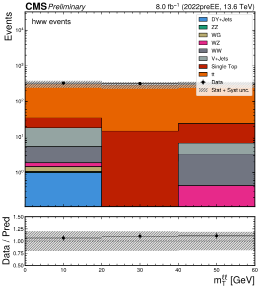


2023postBPix

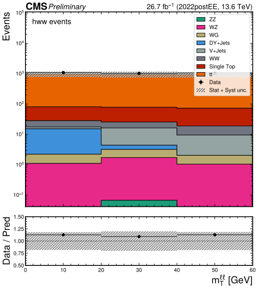


Top Control Region: `mtll`

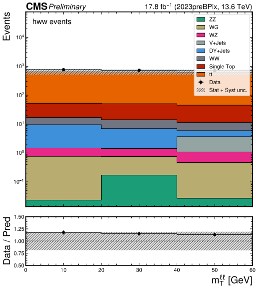
2022preEE



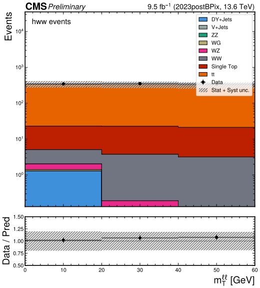
2022postEE



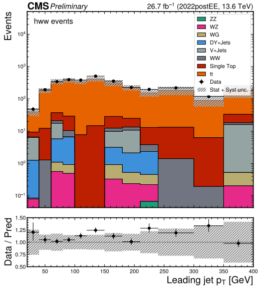
2023preBPix

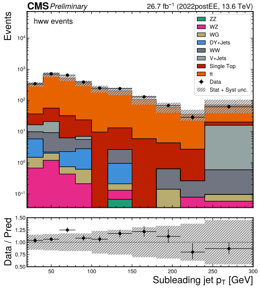


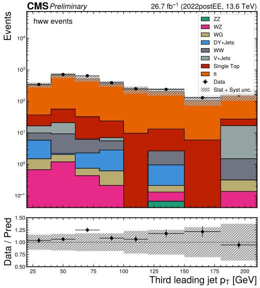
2023postBPix

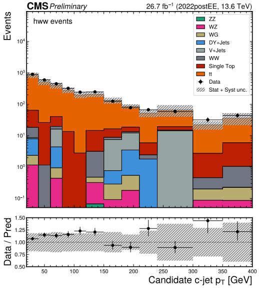


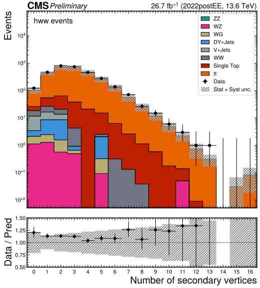
Top CR 2022postEE - extra MVA variables

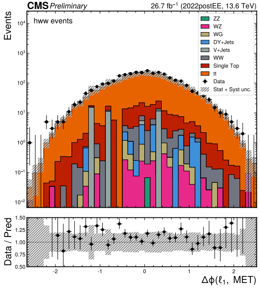


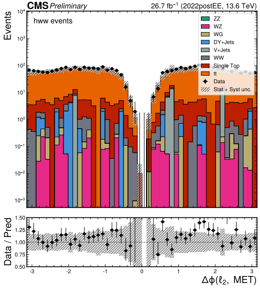


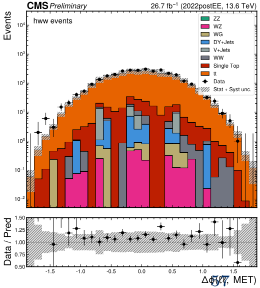




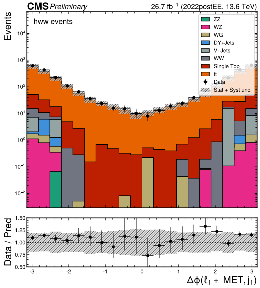


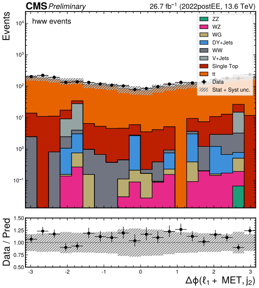


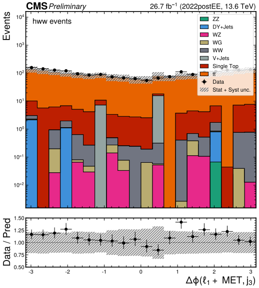


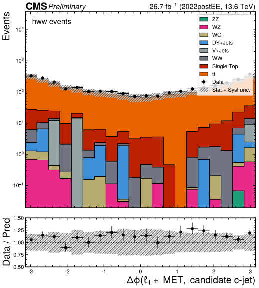


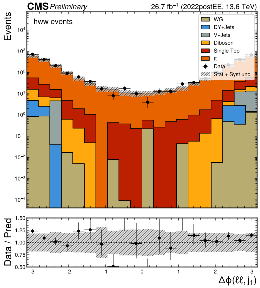
Top CR 2022postEE - extra MVA variables

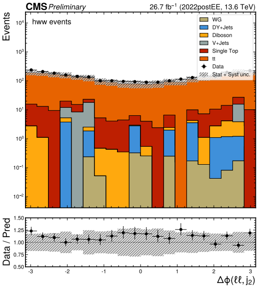


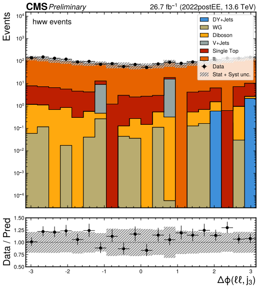


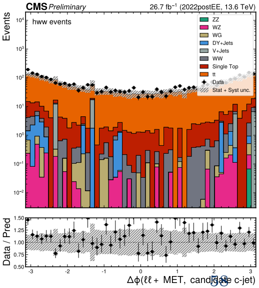




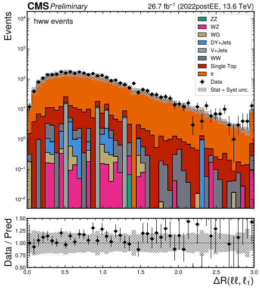


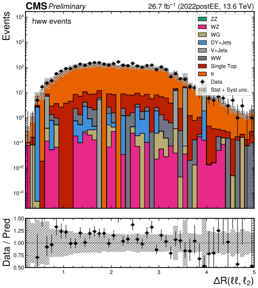


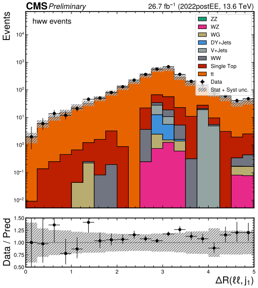


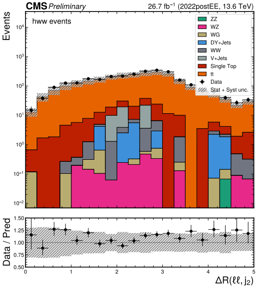


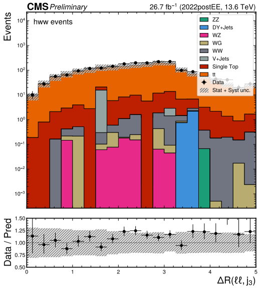
Top CR 2022postEE - extra MVA variables

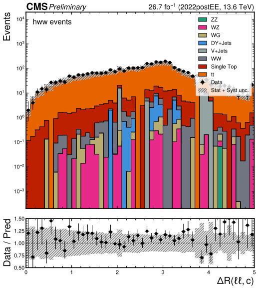






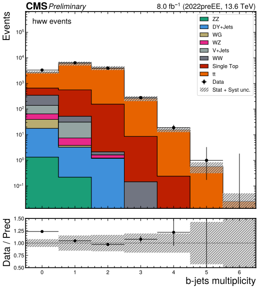




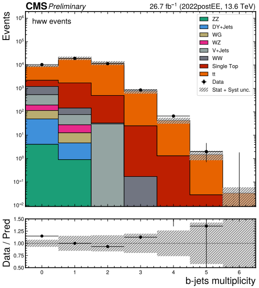


High Mass Control Region: `bjet_multiplicity`

2022preEE



2022postEE



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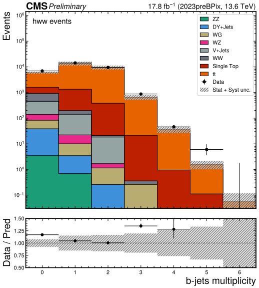

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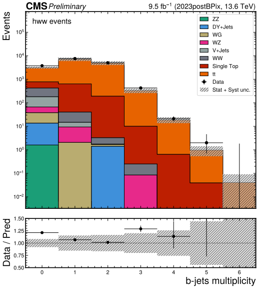

High Mass Control Region: `cjet_cand_cvsb_pnet`

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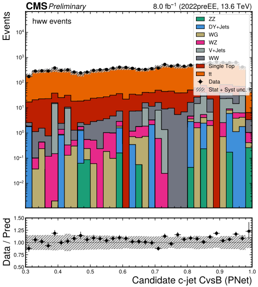

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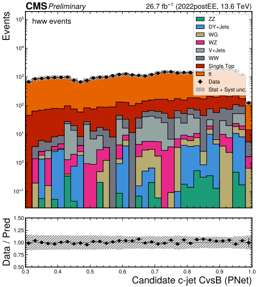

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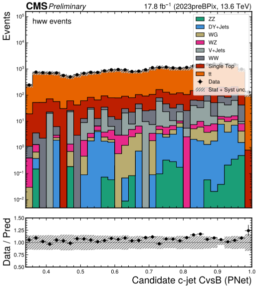

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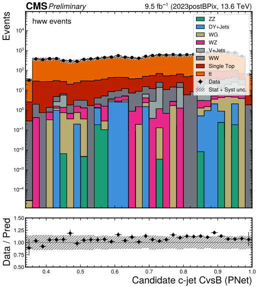

High Mass Control Region: cjet_cand_cvsl_pnet

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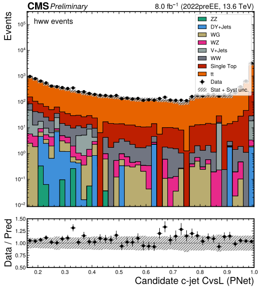

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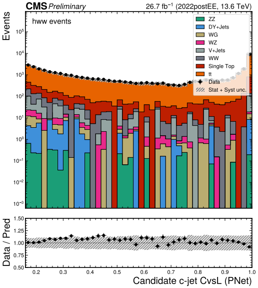

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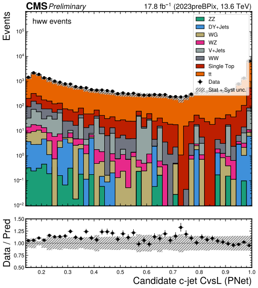

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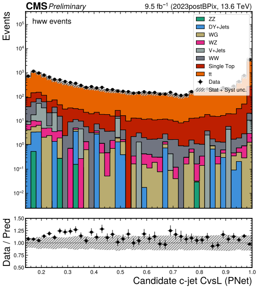
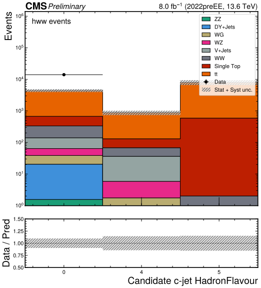


High Mass Control Region: `cjet_cand_flavour`

2022preEE



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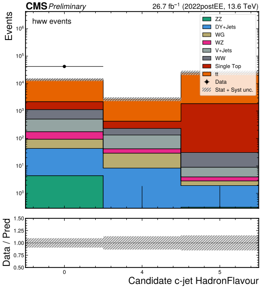

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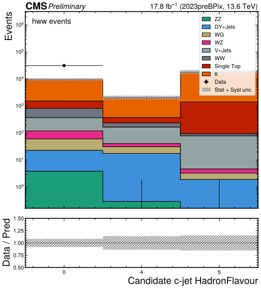

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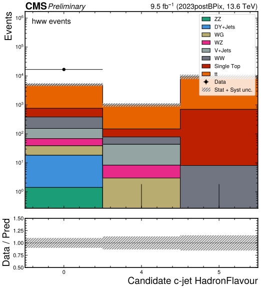
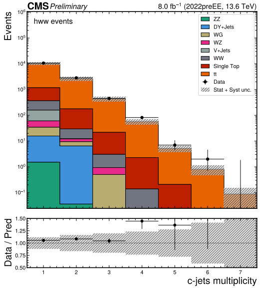


High Mass Control Region: `cjet_multiplicity`

2022preEE



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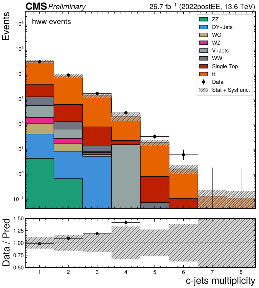

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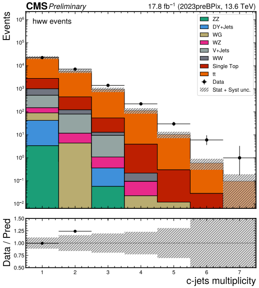

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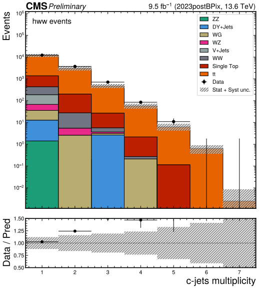

High Mass Control Region: dilepton_mass

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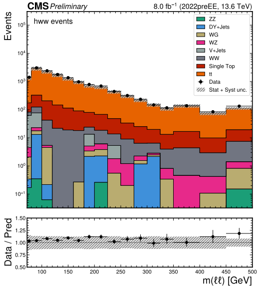

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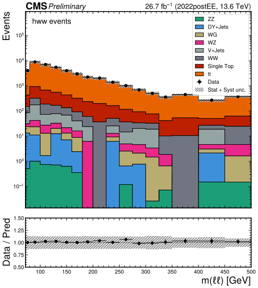

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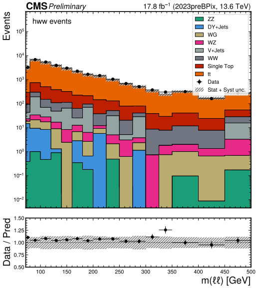

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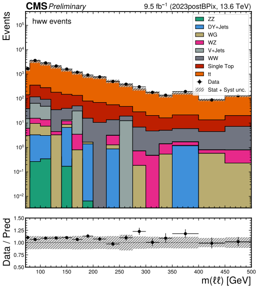

High Mass Control Region: dilepton_pt

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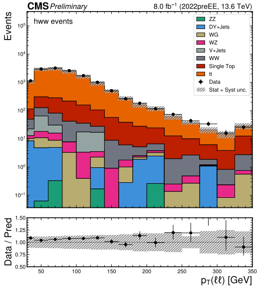

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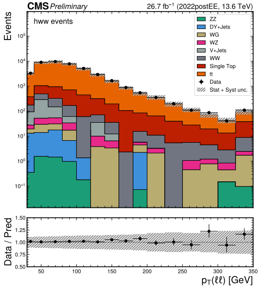

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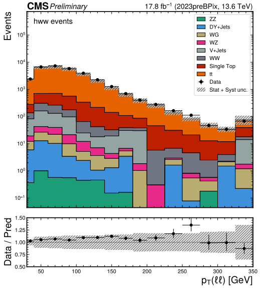

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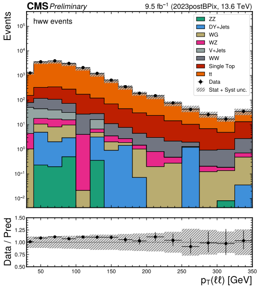
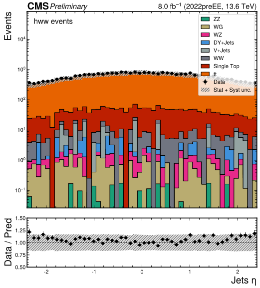


High Mass Control Region: `jet_eta`

2022preEE



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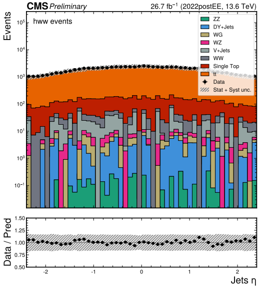

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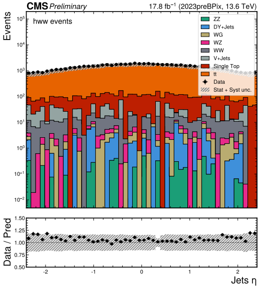

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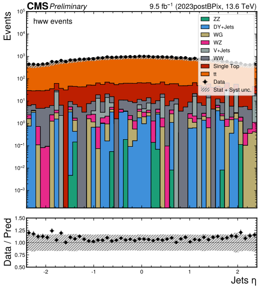
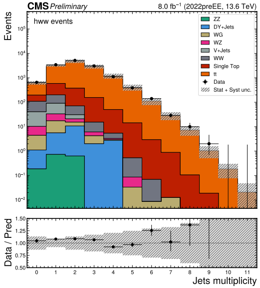


High Mass Control Region: `jet_multiplicity`

2022preEE



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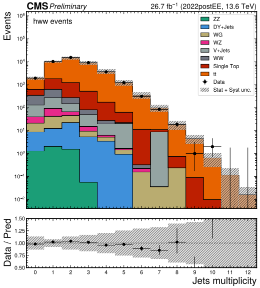

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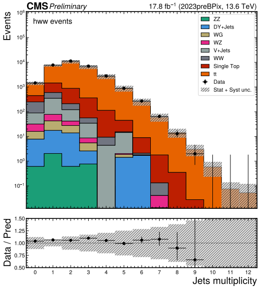

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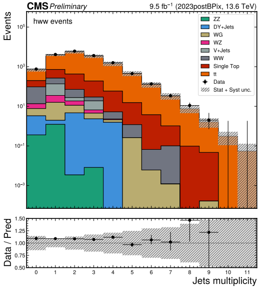
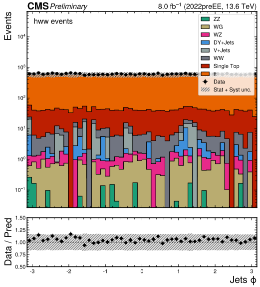
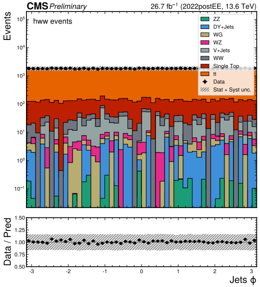


High Mass Control Region: jet_phi

2022preEE



2022postEE



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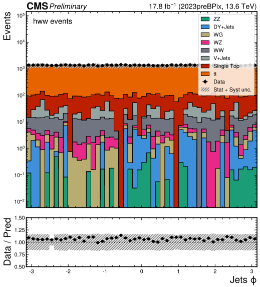

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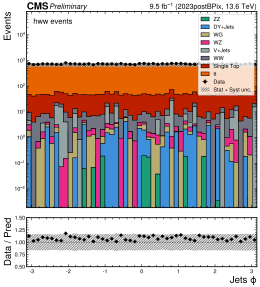
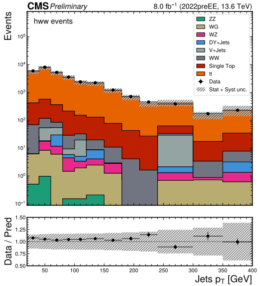


High Mass Control Region: `jet_pt`

2022preEE



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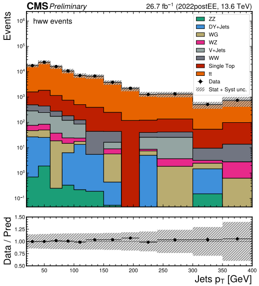

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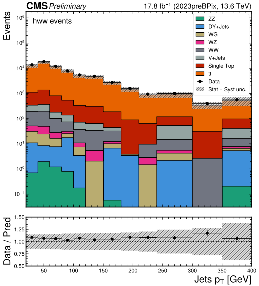

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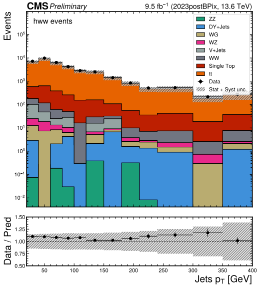
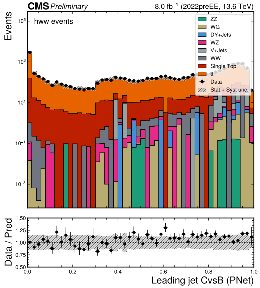
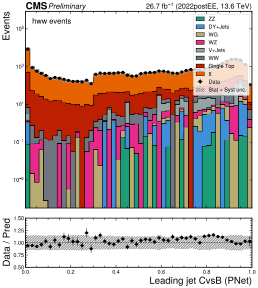


High Mass Control Region:
`leadingjet_cvsb_pnet`

2022preEE



2022postEE



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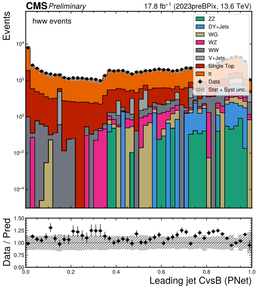

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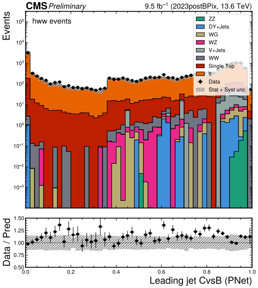
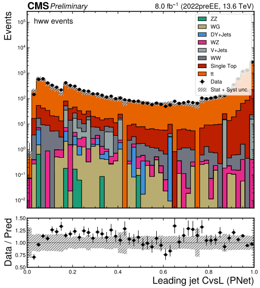


High Mass Control Region: leadingjet_cvsl_pnet

2022preEE



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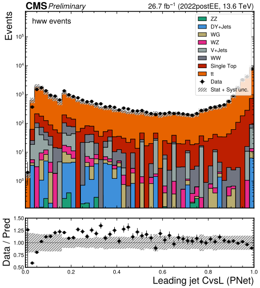

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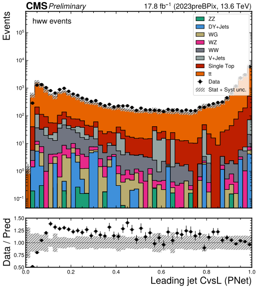

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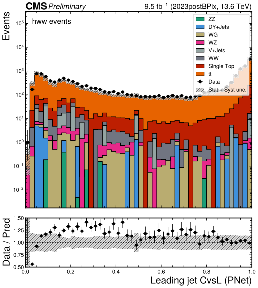
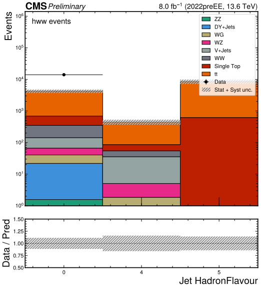
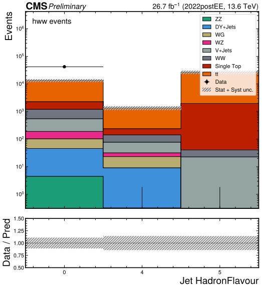


High Mass Control Region: `leadingjet_flavour`

2022preEE



2022postEE



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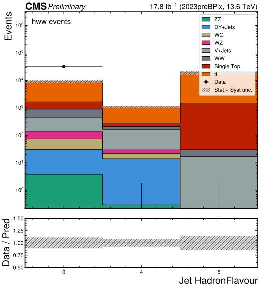

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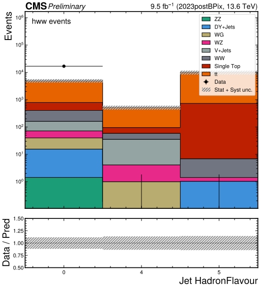

High Mass Control Region: `lepton1_pt`

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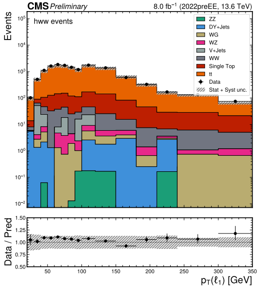

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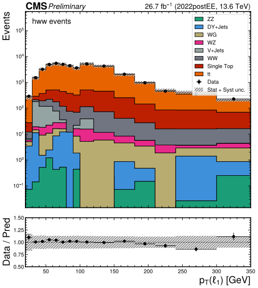

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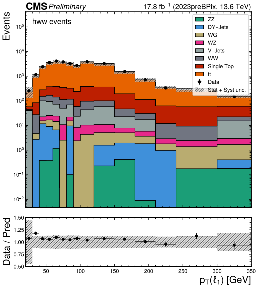

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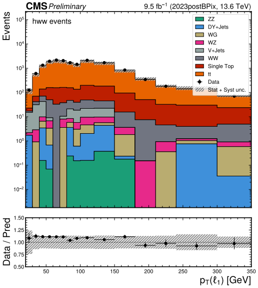

High Mass Control Region: `lepton2_pt`

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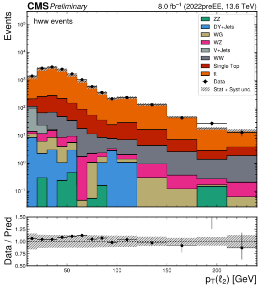

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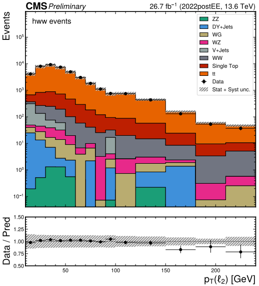

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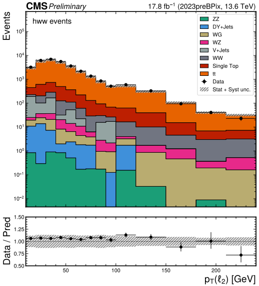

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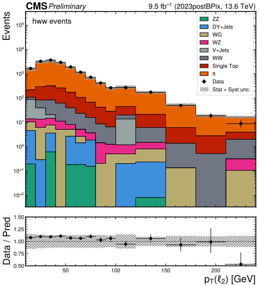

High Mass Control Region: `met_phi`

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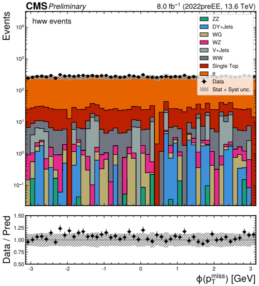

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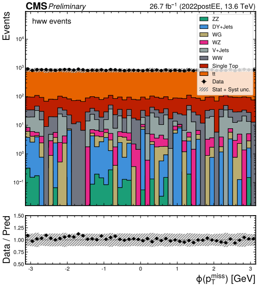

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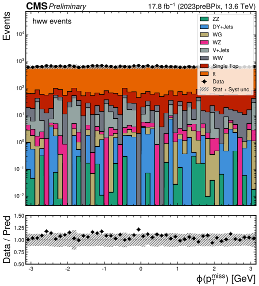

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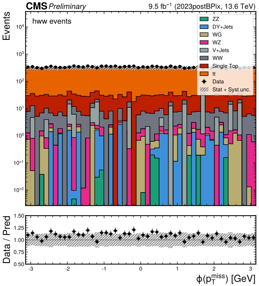

High Mass Control Region: `met_pt`

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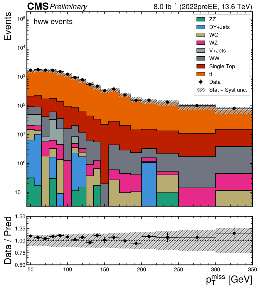

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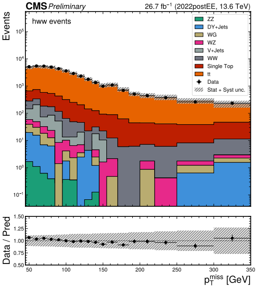

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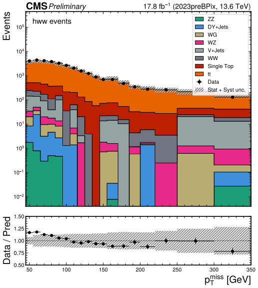

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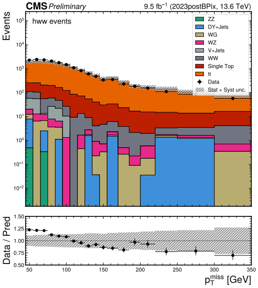

High Mass Control Region: `mtl1`

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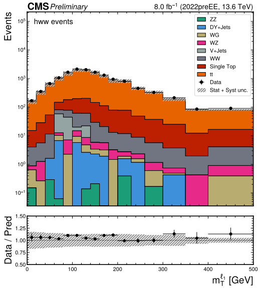

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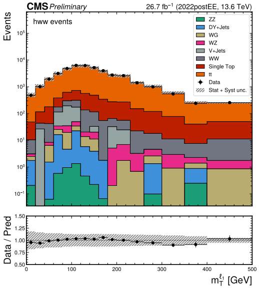

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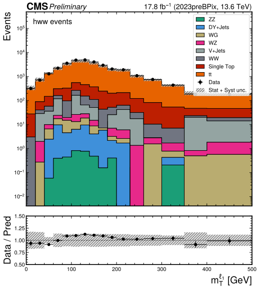

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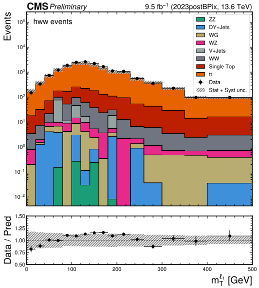

High Mass Control Region: `mtl2`

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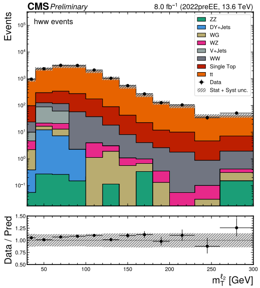

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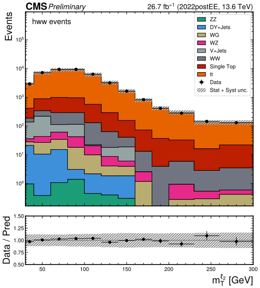

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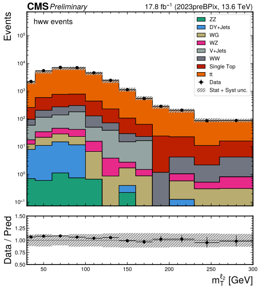

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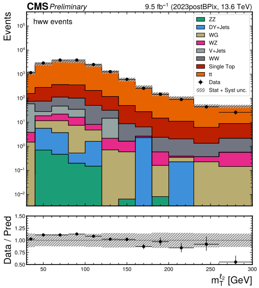

High Mass Control Region: `mtll`

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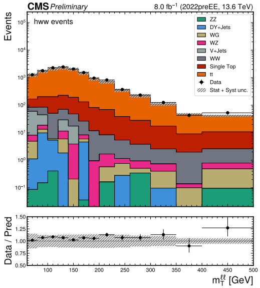

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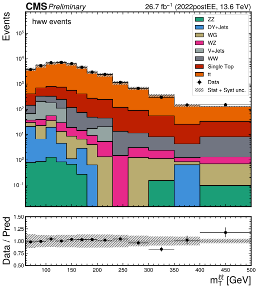

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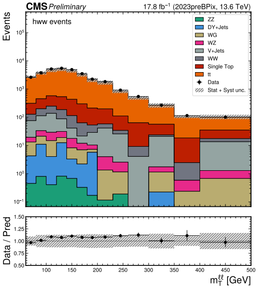

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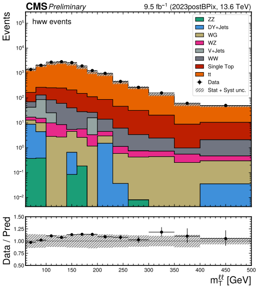
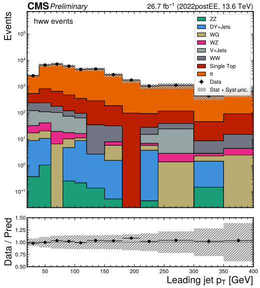
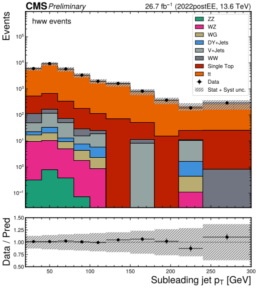
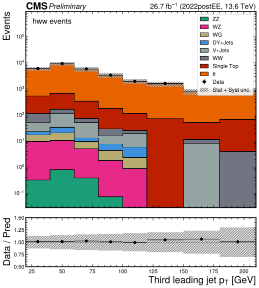
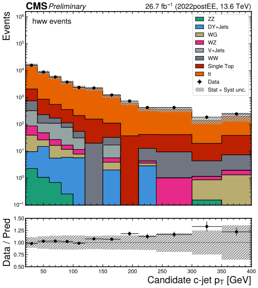


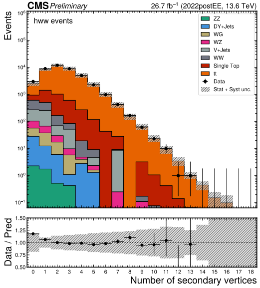
High M CR 2022postEE - extra MVA variables

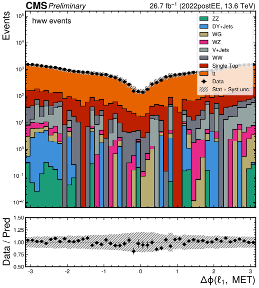


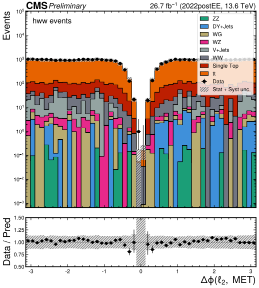


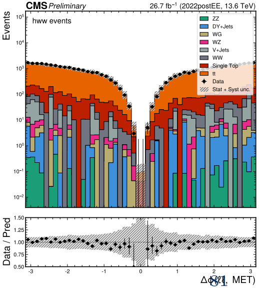




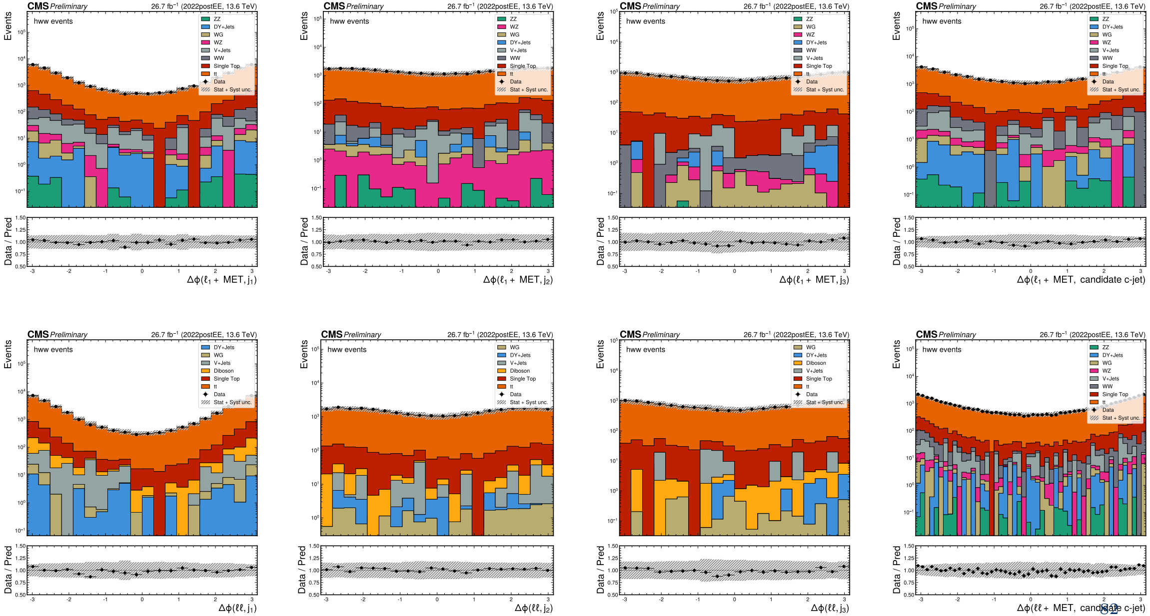








High M CR 2022postEE - extra MVA variables



High M CR 2022postEE - extra MVA variables